

RoadDNA: AI-Powered Crowdsourced Road Health Monitoring System

Abstract

RoadDNA is an intelligent road infrastructure monitoring platform that leverages smartphone sensors, GPS, artificial intelligence, and geospatial analytics to detect potholes and road damage automatically. Instead of relying on manual inspections or citizen complaints, the system continuously collects anonymized sensor data from participating vehicles to generate real-time road health maps and prioritize repairs for municipal authorities.

Problem Statement

Develop a scalable AI-powered platform capable of detecting potholes, road cracks, uneven surfaces, and other road anomalies using smartphone accelerometer, gyroscope, and GPS data. The system should classify road conditions, cluster reports from multiple users, generate severity scores, and provide municipalities with actionable repair priorities through interactive dashboards.

Background

Road maintenance in many cities is reactive, depending on inspections and complaints. Delayed repairs increase accident risks, vehicle maintenance costs, and traffic congestion. With smartphones already containing high-quality motion sensors and GPS, they can serve as a distributed sensing network for continuous road monitoring.

Objectives

1. Collect anonymized accelerometer, gyroscope, speed and GPS data from vehicles.
2. Detect abnormal road events using machine learning.
3. Remove false positives caused by speed breakers or sudden braking.
4. Cluster reports geographically to identify genuine road defects.
5. Generate live road health heatmaps.
6. Prioritize repairs based on severity, traffic density and frequency.
7. Provide dashboards for municipal engineers and administrators.

Proposed System

The mobile application records sensor events while users travel. Suspicious events are transmitted to the backend where AI models classify road anomalies. Confirmed reports are geospatially clustered to eliminate duplicates. The processed data is visualized through a web dashboard showing road quality, hotspot analysis, repair queues and historical trends.

Core Features

- Passive road monitoring using smartphone sensors.
- AI-based anomaly detection.
- GPS-based geospatial clustering.
- Interactive city heatmaps.
- Road health scoring.
- Municipality dashboard.
- Citizen reporting with optional photos.
- Offline data collection and synchronization.
- REST APIs for third-party integration.

Technology Stack

Frontend: Next.js, TypeScript, Tailwind CSS, Mapbox/Leaflet.

Mobile: Flutter (or Android Kotlin) with Accelerometer, Gyroscope and GPS APIs.

Backend: Node.js, Express.js, PostgreSQL, PostGIS, Redis, BullMQ.

AI/ML: Python, FastAPI, Scikit-learn, XGBoost, TensorFlow/PyTorch (future), GeoPandas.

Deployment: Docker, Nginx, Kubernetes (optional).

System Modules

1. Mobile Data Collection Module.
2. Sensor Processing Module.
3. AI Road Damage Classifier.
4. Geospatial Clustering Engine.
5. Heatmap Generation Module.
6. Municipality Dashboard.
7. Analytics & Reporting Module.
8. Notification System.

Innovation

RoadDNA transforms ordinary smartphones into a crowdsourced sensing network capable of continuously monitoring road conditions. Unlike traditional complaint-based systems, it automatically detects road damage, validates reports using crowd consensus and AI, and recommends repair priorities in real time.

Expected Deliverables

- Mobile application.
- AI classification service.
- REST APIs.
- Municipality dashboard.
- Interactive heatmaps.
- Analytics reports.
- Demonstration dataset and deployment guide.

Evaluation Metrics

Detection Accuracy, False Positive Rate, Clustering Precision, API Response Time, Dashboard Performance, Scalability, Explainability of AI predictions, User Experience and Deployment Readiness.

Business Potential

The platform can be offered as a subscription service to municipalities, highway authorities, smart city projects, infrastructure agencies and private road operators. Future offerings include analytics APIs, predictive maintenance services and insurance partnerships.

Future Scope

Integration with dashcams, vehicle telematics and IoT devices; predictive road degradation models; automated work-order generation; drone validation; integration with Smart City platforms and traffic management systems.